Week 8 Notes & Quiz

Notes for all five Week 8 lectures, made only from the 130 slides in *WEEK 8.pdf*. Each yellow tag such as S46 is the slide (PDF page) the point comes from. The 50-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L37	**Strategic Sourcing**	S1–S30
L38	**Vendor Rating**	S31–S51
L39	**Supply Chain Coordination**	S52–S77
L40	**Supply Chain Performance: Balanced Scorecard Approach**	S78–S98
L41	**Supply Chain Performance: Balanced Scorecard Approach (contd.)**	S99–S130
+	**Cheat sheet · Slide quirks · Quiz (50 Qs)**	—

LECTURE 37 S1–S30

Strategic Sourcing

STRATEGIC SOURCING S3

A procurement strategy for identifying, evaluating and selecting suppliers on specific criteria to optimize the supply chain and meet long-term goals. It goes **beyond finding the lowest-cost supplier**: build strong relationships, improve quality and delivery, reduce cost and risk, and align sourcing with organizational strategy.

Characteristics 7 items

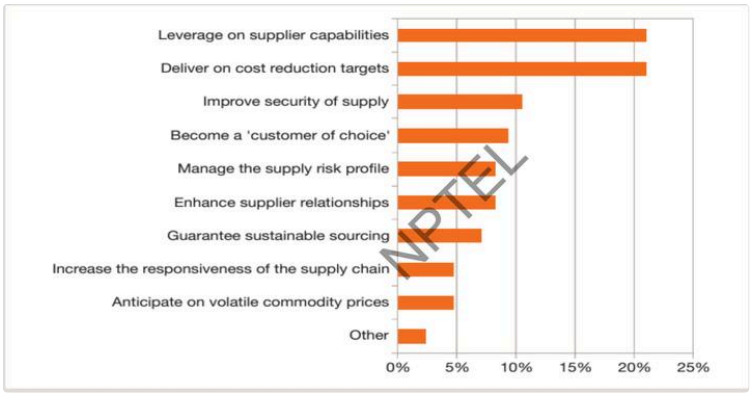
Disciplined buying practices · **total cost focus** (total cost of ownership, not just price) · consolidated purchasing power · tight supplier relationships · improved teamwork and purchasing skills · realigned business processes · continuous improvement · focus on value creation. S4–S6

Supplier selection and management

- Suppliers matter for meeting requirements, controlling unit cost, cutting total acquisition cost, relying on **fewer suppliers**, and designing subassemblies. Suppliers are treated as **partners**. S7
- Selection criteria: price, quality, delivery, performance history, warranties and claims, capacity, technical capability, financial position. S8
- **Supplier certification**: certified suppliers have proven they consistently deliver quality, which **reduces routine testing**. S8

Supplier Relationship Management (SRM)

- Collaborative, mutually beneficial relationships with key suppliers based on trust and a **win-win** orientation. Objectives include becoming the **"customer of choice"**, leveraging supplier capabilities for innovation, sharing growth, profit, risk and investment. S10
- Setting up SRM: define the strategy → identify key suppliers → assess capabilities → governance structure (cross-functional) → collaboration initiatives → invest in technology → open communication → performance measurement → supplier development programs. S11–S13



Original slide S14 (PwC). Top SRM objectives: **leverage supplier capabilities** and **deliver on cost reduction targets** (about 21% each), then security of supply and becoming the customer of choice. Top challenge on S15: **too much focus on costs instead of value**.

Supplier segmentation

KRALJIC SEGMENT	PROFIT IMPACT · SUPPLY RISK	HOW TO MANAGE S16 S19
Strategic	High · High	Close partnerships, joint development, executive involvement
Leverage	High · Low	Cost optimization, competitive bidding, use buying power
Bottleneck	Low · High	Ensure continuity: safety stock, dual sourcing, alternative suppliers

KRALJIC SEGMENT	PROFIT IMPACT · SUPPLY RISK	HOW TO MANAGE S16 S19
Non-critical	Low · Low	Efficient transactions, automation, minimal resources

- Segmentation criteria: spend, risk, criticality, performance, market position, geographic location. S17
- **ABC analysis** (Pareto 80/20): A = few suppliers, most spend; B = moderate; C = many suppliers, low spend. S18

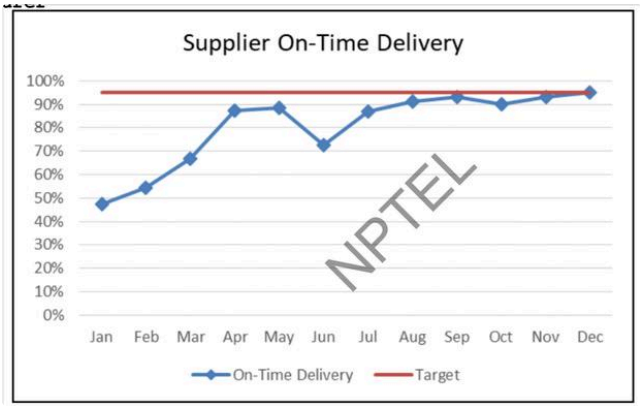
Global sourcing: eight dimensions

Cost (total landed cost) · quality · CSR · risk · opportunities · politics · local context · enabling infrastructure. S20–S21

Case: Colorado manufacturer

- Founded 1950s; build-to-order, low-volume high-mix; outdated MRP → about **$1M** excess and obsolete inventory; customer lead time **12 weeks**, double competitors'. S22–S23
- New President (Lean expert) spent the first **100 days** assessing; consultant partly funded by the Rocky Mountain Trade Adjustment Assistance Center; cross-functional team. S23
- Seven steps: define objectives (cut lead time **80%** to two weeks) → specify critical parts (machined parts; total landed cost; local suppliers) → assess suppliers (NDA, questionnaire, 3-year price stability) → select and allocate → award to **five shops** (10–30% each), Kanban → implement (blanket PO, new ERP, daily buyer walk) → track (quarterly scorecard). S24–S25

RESULT	BEFORE → AFTER S26
Machined part lead time	6–8 weeks → 2 days
Supplier on-time delivery	50% → 95% (first year)
Supplier quality	60% → over 99%
Gross profit	+30%
Inventory	−50% ($750,000 freed)
Customer lead time	12 → 2 weeks (−80%)
Quality · warranty	+25% · 0.2% of sales
Market share	+30% over four years



Original slide S27: supplier on-time delivery, January to December, rising to the target line.

LECTURE 38 S31–S51

Vendor Rating

- **Vendor rating**: a formal evaluation system that scores suppliers on criteria such as delivery, quality, price and service; often a weighted formula of quality and delivery ratings. Important where large sums are involved, e.g. **railways**. S33
- Four core criteria: **quality, price, delivery, service**; also reliability, innovation, relationship. S34–S36
- Need: performance evaluation, supplier development, risk mitigation, cost optimization, benchmarking, supplier relationship management. S37–S38

8 steps S39–S41: define criteria → assign weights → gather data → evaluate performance → calculate ratings (e.g. weighted scoring) → review and validate → communicate and give feedback → use for decisions.

Techniques

Categorical plan	Managers list factors from experience and grade vendors **good, satisfactory, poor**; **very subjective**. S43
Weighted point plan	Factors are weighted and vendors scored on each. S43
Cost ratio plan	Cost of quality, price, delivery etc. as a % of total purchase value. Example: delivery cost Rs 5,000 on purchases of Rs 1,00,000. S44
Forced decision matrix	Attributes compared in pairs: the more important gets 1, the other 0. S45
Service cost ratio	Subjective intangibles: labour stability, financial stability, flexibility for rush orders, R&D. S45
IBM quality rating	**VGR = desired cost of inspection ÷ actual cost of inspection × 100**. S45
Bell quality rating	Bell Helicopter's **Lot Quality Index**: LQI = X ÷ L, with X = L1×1.00 + L2×2.10 + L3×2.90 + L4×3.10 + L5×3.90. S46

- Bell lots: L1 accepted as received; L2 rejected by sampling but labelled; L3 rejected, reworked at supplier; L4 rejected, returned not usable; L5 rejected, reworked at Bell. L = total lots received. S46
- The technique chart also lists "Eavaston's vendor selection". S42
- Advantages: less subjectivity, feedback from all areas in one package, better communication, overall control of the vendor base, specific corrective action, continuous review standards. S48

LECTURE 39 S52–S77

Supply Chain Coordination

- **Coordination**: strategic alignment of processes, activities and information among suppliers, manufacturers, distributors and retailers to optimize the whole chain. Key elements: demand forecasting, inventory, production planning, transport, communication. S54
- With suppliers: collaborative forecasting, information sharing, **VMI**, **JIT delivery**, quality standards, regular meetings, integrated IT. S55–S56
- With distributors: efficient order processing, distribution planning, stock replenishment, safety stock, reliable delivery, returns, sales data sharing, KPIs, joint business planning. S57–S58
- Coordination points between suppliers and distributors: purchase orders, production plan, inventory management, delivery and transportation. S59–S60
- Lack of coordination causes the **bullwhip effect**. Causes: forecasting errors, **order batching**, **price fluctuations**, lack of information sharing. Effects: higher costs, excess inventory, stockouts, poor service. S61 S63

PERFORMANCE MEASURE	IMPACT OF LACK OF COORDINATION S64
Manufacturing, inventory, transportation, shipping and receiving cost	**Increase**
Replenishment lead time	**Increases**
Level of product availability	**Decreases**
Profitability	**Decreases**

Obstacles to coordination

Incentive (inventive) obstacles · information processing obstacles · operational obstacles · pricing obstacles · behavioural obstacles (lack of trust). S62

Managerial actions 5 groups

1. Align goals and incentives	Maximize total SC surplus; pricing for coordination (lot-size quantity discounts, two-part tariffs, buyback, revenue-sharing, quantity flexibility); sales force incentives from **sell-in to sell-through**. S65–S66
2. Information visibility and accuracy	Share **POS data**; collaborative forecasting; **single-stage control** of replenishment (CRP, VMI). S67
3. Operational performance	Reduce replenishment lead time; reduce lot sizes; ration based on **past sales** to limit gaming. S68
4. Pricing to stabilize orders	Move from lot-size-based to **volume-based** discounts; **EDLP**; tie promotions to sell-through. S69
5. Strategic partnerships and trust	Trust and information sharing; clear contracts and conflict resolution. S70

- Macro processes: CRM, ISCM, SRM. S71
- **Amazon CRM**: personalized recommendations, **one-click ordering**, order visibility, call-back option. 2008 CRM market (Gartner): **SAP 22.5%**, Oracle 16.1%, Salesforce.com 10.6%, Microsoft 6.4%, Amdocs 4.9%. S72–S73
- Hindustan Oil Company (HOC): efficient SCM, quality, R&D, sustainability. S74

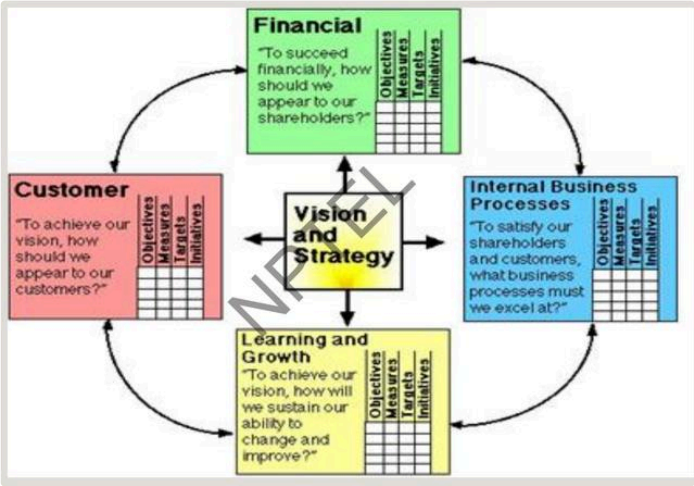
LECTURE 40 S78–S98

Supply Chain Performance: Balanced Scorecard

- Measures: **qualitative** (customer satisfaction, product quality) and **quantitative** (order-to-delivery lead time, response time, flexibility, resource utilization). Quantitative splits into financial and non-financial. S80–S81
- Financial measures: raw material cost, revenue, holding cost, transport cost, activity-based costs, expired perishables, penalties and credits for late or wrong orders, returns. S82
- Non-financial: **cycle time** (supply chain lead time; order-to-delivery lead time), customer service level (**order fill rate**, **stockout rate** = its reverse, backorder level, probability of on-time delivery), inventory (raw, WIP, spares, finished), resource utilization (manufacturing, storage, logistics, human, financial). S83–S85
- Qualitative: satisfaction, complaints, response time, quality, delivery experience, order accuracy. S86

BALANCED SCORECARD S87

Developed by **Robert Kaplan and David Norton**. Combines financial and non-financial indicators across four perspectives: **Financial, Customer, Internal Business Processes, Learning and Growth**, aligning activities with vision and strategy.



Original slide S88 (Kaplan & Norton). Financial: how should we appear to shareholders? Customer: how should we appear to customers? Internal processes: what must we excel at? Learning and growth: how will we sustain our ability to change and improve? Each has objectives, measures, targets, initiatives.

Customer	Cost, service, quality, time: "Number One in delivering value to customers". S89
Internal process	Activities per function, bottlenecks, alignment, automation. S90
Learning & growth	Job satisfaction, turnover, training, right expertise. S91
Financial	ROI, cash flow, quarterly/annual results, ROCE. S92

- Benefits: aligns measures with strategy at all levels, turns the mission into daily "marching orders"; used by nearly 60% of Fortune 500 companies. S95
- Pitfalls: no well-defined strategy, too much focus on **lagging** measures, generic copied metrics, self-serving managers. S96

LECTURE 41 S99–S130

Balanced Scorecard (contd.): QDC Indices and Techniques

- SC BSC: Finance (revenue, gross margin, ROI), Clients (satisfaction, loyalty), Internal (machine productivity, response times), Improvement & innovation (training, new products). S101–S104
- Procurement BSC: financial (savings, TCO, cost per transaction), customer (complaints, internal satisfaction), internal processes (suppliers for 90% of spend, lead time from requisition to order), learning (training, certification, turnover). S105
- Implementing: define objectives → develop metrics → collect and analyse data → review and adjust → foster collaboration. S106

QDC performance indices

INDEX	FORMULA
Quality (QPIs) S109	
Percentage of rejection	Defects received ÷ items received × 100
DPMO	Defects found ÷ items tested × 10^6
Supplier rejection %	Number of rejections ÷ total suppliers × 100
% non-compliance	Non-compliances ÷ total observations × 100
Delivery (DPIs) S110	
Supplier OTIF	POs fulfilled on time and in full ÷ POs placed × 100
Delivery in full, on time	% in full × % on time – % rejected
Supplier lead time variability	Actual delivery time – lead time
Overall lead time variability	Actual delivery time – required lead time
% shipments in good condition	Non-defective shipments ÷ shipments received × 100
Cost (CPIs) S111–S112	
Cost Performance Index	**CPI = Earned Value ÷ Actual Cost**; EV = Planned Value × % complete. CPI = 1 on budget, >1 under budget, <1 over budget
Cost avoidance	Actual purchasing price – lowest price quoted
Cost reduction	Actual purchasing price – lowest price paid
Procurement ROI	(Cost reduction + cost avoidance) ÷ cost of procurement operations
Procurement cost % of sales	Total procurement cost ÷ total sales × 100

16 measurement techniques

1–2. Inventory investment, efficiency	Investment as % of revenue; efficiency via inventory turns and **days of supply**; exclude stock with no demand in 30/60/90 days. S113–S114
3. On-time supplier delivery	On-time %, actual hours late or early. S115
4. Forecast accuracy	Average deviation, **average absolute deviation** (stops errors cancelling), **root mean squared deviation** (for safety stock). S116
5–10	Lead time, unplanned orders, schedule changes, overdue backlog, material availability, excess and obsolete inventory (Lean Six Sigma tools: VSM, quick changeover, 5S, TPM, mistake-proofing). S117–S119
11–12. Service targets, perfect order	**Perfect order = order entry accuracy × pick accuracy × on-time delivery × damage-free shipping × invoice accuracy**. S120
13–16. Financial	Gross profit margin (revenue – COGS); asset efficiency (sales ÷ average assets); **ROA = net profit margin × asset efficiency**; **GMROI** = gross margin ÷ average inventory investment. S121–S122

- **ShipBob** (tech-enabled 3PL): tracks perfect order rate, costs, fill rate, lead times, cash-to-cash cycle. S123–S124
- **Southwest Airlines**: a scorecard linking ground-crew metrics like "fast ground turnaround" to "lower costs"; detail for supervisors, a dashboard for top management (Mike Van de Ven). S125–S126

BEFORE THE EXAM ALL LECTURES

Cheat sheet

Kraljic	Strategic (H,H) · leverage (H profit, L risk) · bottleneck (L profit, H risk) · non-critical (L,L) S16
Vendor rating criteria	Quality, price, delivery, service S34
IBM VGR	Desired ÷ actual cost of inspection × 100 S45
Bell LQI	X ÷ L; weights 1.00, 2.10, 2.90, 3.10, 3.90 S46
Bullwhip causes	Forecast errors, order batching, price fluctuations, no information sharing S61
BSC	Kaplan & Norton: financial, customer, internal processes, learning & growth S87
DPMO	Defects ÷ items tested × 10^6 S109
CPI	EV ÷ AC; EV = PV × % complete S111
Perfect order	Entry × pick × on-time × damage-free × invoice accuracy S120
ROA	Net profit margin × asset efficiency S122

NUMBER	MEANING	SLIDE
`12 → 2 weeks`	Colorado customer lead time (−80%)	S26
`50% → 95%`	Colorado supplier on-time delivery	S26
`$750,000`	Cash freed by 50% inventory cut	S26
`22.5%`	SAP's CRM market share (2008)	S73
`~60%`	Fortune 500 using some BSC	S95

WORKED EXAMPLE (MY OWN NUMBERS, USING THE SLIDE FORMULAS)

Planned value ₹1,00,000, 60% complete, actual cost ₹50,000 → EV = ₹60,000; CPI = 60,000 ÷ 50,000 = **1.2** (under budget). Bell LQI with 10 lots: 6 accepted, 2 L2, 1 L3, 1 L4 → X = 6 + 4.2 + 2.9 + 3.1 = 16.2 → LQI = **1.62** (closer to 1 is better).

READ ONCE SOURCE CHECK

Slide quirks to watch

- **Cost ratio example:** Rs 5,000 ÷ Rs 1,00,000 × 100 = 5%, but the slide prints "10%". S44
- **Inventory turns and DOS formulas** are garbled on the slide ("Annualized COGS × Average Inventory Investment", "Inventory Investment × Daily COGS"). The usual forms are turns = annual COGS ÷ average inventory and DOS = inventory ÷ daily COGS. S114
- "% Non-compliance rectified" divides by "total number of non-compliance rectified"; the denominator should be total non-compliances. S109
- "Inventive obstacles" in the coordination slide is Chopra's "incentive obstacles". S62
- "Eavaston's vendor selection" appears in the technique chart but is never explained; the "Vendor Rating Format" slide (S47) is blank in the PDF.
- The Lecture 37 title slide reads "37: Strategic Sourcing / Lecture –". S1

SELF-TEST 50 QUESTIONS

Week 8 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 37

Strategic sourcing differs from simple purchasing because it:

A goes beyond lowest cost to build relationships, improve quality and delivery, and align with organizational strategy
B only handles global suppliers
C avoids supplier relationships
D always picks the lowest-price supplier

Q2 LECTURE 37

The 'total cost focus' of strategic sourcing means considering:

A only tax
B only transport cost
C the total cost of ownership including quality, service and delivery
D only the purchase price

Q3 LECTURE 37

Supplier certification mainly helps by:

A reducing the need for routine testing because quality is proven
B increasing routine testing
C raising prices
D adding more suppliers

Q4 LECTURE 37

In the Kraljic matrix, suppliers with HIGH profit impact and LOW supply risk are:

A Strategic
B Leverage
C Bottleneck
D Non-critical

Q5 LECTURE 37

Bottleneck suppliers should be managed mainly through:

A executive joint development
B safety stock, dual sourcing or alternative suppliers
C full automation
D competitive bidding

Q6 LECTURE 37

ABC supplier analysis is based on:

A the Kraljic matrix
B the bullwhip effect
C the Pareto principle (80/20 rule)
D Little's law

Q7 LECTURE 37

An SRM objective is to become the supplier's:

A 4PL
B customer of choice
C only buyer
D franchisee

Q8 LECTURE 37

According to the PwC chart, the top key challenge in SRM is:

A lack of IT
B too much focus on costs instead of value
C too many suppliers
D lack of mutual understanding

Q9 LECTURE 37

In the Colorado manufacturer case, the customer lead time fell from:

A 6–8 weeks to 2 days
B 12 weeks to 2 weeks
C 100 days to 10 days
D 8 weeks to 4 weeks

Q10 LECTURE 37

Supplier on-time delivery at the Colorado manufacturer improved in the first year from:

A 50% to 95%
B 30% to 80%
C 60% to 99%
D 80% to 100%

Q11 LECTURE 37

Which is one of the eight dimensions of the global sourcing framework?

A Stock price
B Brand colour
C Advertising reach
D Enabling infrastructure

Q12 LECTURE 38

The four core criteria for vendor rating shown on the slide are:

A cost, time, risk, scale
B plan, source, make, deliver
C quality, price, delivery, service
D volume, value, velocity, variety

Q13 LECTURE 38

Which vendor rating technique grades vendors as good, satisfactory or poor and is very subjective?

A Weighted point plan
B Categorical plan
C Cost ratio plan
D Forced decision matrix

Q14 LECTURE 38

In the forced decision matrix, when one attribute is more important than another, it is assigned:

A a weight of 10 and the other 5
B a weight of 1 and the other 0
C equal weights
D no weight

Q15 LECTURE 38

The IBM quality rating system computes VGR as:

A defects ÷ items × 10^6
B lots rejected ÷ lots received
C desired cost of inspection ÷ actual cost of inspection × 100
D actual ÷ desired cost of inspection × 100

Q16 LECTURE 38

The Bell quality rating system's Lot Quality Index is calculated as:

A LQI = L ÷ X
B LQI = L1 ÷ L
C LQI = X ÷ L
D LQI = X × L

Q17 LECTURE 38

In the Bell system, which lot category carries the weight 1.00?

A Lots rejected by sampling but labelled
B Lots acceptable as received
C Lots rejected and returned not usable
D Lots reworked at Bell

Q18 LECTURE 38

Using the Bell formula, 10 lots with 6 accepted (L1), 2 in L2, 1 in L3 and 1 in L4 give an LQI of:

A 2.10
B 16.2
C 1.62
D 1.00

Q19 LECTURE 38

Which technique subjectively measures intangibles like labour stability, financial stability and flexibility for rush orders?

A Bell quality rating
B Service cost ratio
C IBM quality rating
D Categorical plan

Q20 LECTURE 38

What is the second step of vendor rating?

A Gather data
B Assign weights to criteria
C Communicate results
D Calculate ratings

Q21 LECTURE 38

Vendor rating is described as especially important in which industry on the slide?

A Fashion retail
B Hospitality
C Software
D Railways

Q22 LECTURE 39

Which is NOT listed as a cause of the bullwhip effect?

A Sharing POS data
B Demand forecasting inaccuracies
C Order batching
D Price fluctuations

Q23 LECTURE 39

According to the table on lack of coordination, product availability:

- **A** decreases
- **B** doubles
- **C** stays the same
- **D** increases

Q24 LECTURE 39

Lack of coordination causes replenishment lead time to:

- **A** stay constant
- **B** vanish
- **C** increase
- **D** decrease

Q25 LECTURE 39

Shifting sales force incentives from 'sell-in' to 'sell-through' helps by:

- **A** reducing forward buying and order fluctuations
- **B** raising prices
- **C** shortening transit
- **D** encouraging forward buying

Q26 LECTURE 39

Continuous replenishment programs (CRP) and VMI are examples of:

- **A** sell-in incentives
- **B** order batching
- **C** lot-size discounts
- **D** single-stage control of replenishment

Q27 LECTURE 39

During shortages, rationing based on ____ discourages artificial order inflation.

- **A** current orders
- **B** price
- **C** distance
- **D** past sales

Q28 LECTURE 39

To stabilize orders, the slides recommend moving from lot-size-based discounts to:

- **A** no discounts at all
- **B** volume-based quantity discounts
- **C** high-low promotions
- **D** lot-for-lot pricing

Q29 LECTURE 39

Lack of trust is classified as which obstacle to coordination?

- **A** Behavioural
- **B** Pricing
- **C** Information processing
- **D** Operational

Q30 LECTURE 39

Per Gartner, the largest CRM software provider in 2008 was:

- **A** SAP
- **B** Oracle
- **C** Microsoft
- **D** Salesforce.com

Q31 LECTURE 39

Amazon's feature that lets customers buy without re-entering payment or shipping details is:

- **A** FBA
- **B** one-click ordering
- **C** Scout
- **D** Prime

Q32 LECTURE 40

The Balanced Scorecard was developed by:

- **A** Robert Kaplan and David Norton
- **B** Michael Porter
- **C** Deming and Shewhart
- **D** Chopra and Meindl

Q33 LECTURE 40

Which is NOT one of the four BSC perspectives?

- **A** Customer
- **B** Internal business processes
- **C** Financial
- **D** Supplier bargaining power

Q34 LECTURE 40

Customer demand satisfied directly from available stock is the:

- **A** cycle time
- **B** order fill rate
- **C** backorder level
- **D** stockout rate

Q35 LECTURE 40

Cycle time is often called:

- **A** idle time
- **B** lead time
- **C** dwell time
- **D** takt time

Q36 LECTURE 40

Customer satisfaction and product quality are examples of:

- **A** qualitative measures
- **B** quantitative financial measures
- **C** inventory levels
- **D** resource utilization

Q37 LECTURE 40

Which perspective asks 'To achieve our vision, how will we sustain our ability to change and improve?'

- **A** Financial
- **B** Customer
- **C** Learning and growth
- **D** Internal business process

Q38 LECTURE 40

Roughly what share of Fortune 500 companies use some form of Balanced Scorecard?

- **A** 20%
- **B** 60%
- **C** 90%
- **D** 40%

Q39 LECTURE 40

Which is a criticism or pitfall of the Balanced Scorecard?

- **A** It ignores financial measures
- **B** It has only one perspective
- **C** It cannot be used in supply chains
- **D** Too much focus on lagging measures

Q40 LECTURE 40

In the customer perspective slide, which four dimensions surround 'Number One in delivering value to customers'?

- **A** Bottlenecks, alignment, automation, activities
- **B** Cost, service, quality, time
- **C** Training, turnover, satisfaction, expertise
- **D** ROI, cash flow, ROCE, results

Q41 LECTURE 41

Defects per Million Opportunities (DPMO) is:

- **A** defects found ÷ items tested × 100
- **B** rejections ÷ suppliers × 100
- **C** defects found ÷ items tested × 10^6
- **D** defects ÷ lots

Q42 LECTURE 41

The Cost Performance Index (CPI) equals:

- **A** Earned value ÷ actual cost
- **B** Actual cost ÷ earned value
- **C** Earned value − actual cost
- **D** Planned value ÷ actual cost

Q43 LECTURE 41

A project has planned value ₹1,00,000, is 60% complete, and has spent ₹50,000. Its CPI is:

- **A** 1.2
- **B** 0.83
- **C** 1.0
- **D** 2.0

Q44 LECTURE 41

A CPI less than 1 means the project is:

- **A** on budget
- **B** complete
- **C** over budget
- **D** under budget

Q45 LECTURE 41

Cost avoidance is defined as:

- **A** EV ÷ AC
- **B** actual purchasing price − lowest price quoted
- **C** actual purchasing price − lowest price paid
- **D** total procurement cost ÷ sales

Q46 LECTURE 41

Procurement ROI equals:

- **A** (Cost reduction + cost avoidance) ÷ cost of procurement operations
- **B** sales ÷ procurement cost
- **C** EV ÷ PV
- **D** gross margin ÷ inventory

Q47 LECTURE 41

The perfect order metric multiplies order entry accuracy, pick accuracy, on-time delivery, damage-free shipping and:

A fill rate
B lead time
C invoice accuracy
D inventory turns

Q48 LECTURE 41

Return on Assets (ROA) is calculated on the slides as:

A revenue − COGS
B net profit margin × asset efficiency
C gross margin ÷ inventory
D sales ÷ assets

ANSWERS 50 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	S3. S3
Q2	C	S4. S4
Q3	A	S8. S8
Q4	B	Strategic H/H, leverage H/L, bottleneck L/H, non-critical L/L. S16
Q5	B	Low profit impact but high supply risk: ensure continuity. S19
Q6	C	A-class: few suppliers with most spend. S18
Q7	B	S10. S10
Q8	B	S15. S15
Q9	B	An 80% cut. Machined-part lead time went from 6–8 weeks to 2 days. S26
Q10	A	Supplier quality went from 60% to over 99%. S26
Q11	D	Cost, quality, CSR, risk, opportunities, politics, local context, enabling infrastructure. S21
Q12	C	S34. S34
Q13	B	S43. S43
Q14	B	S45. S45
Q15	C	S45. S45
Q16	C	L = total lots received; X = weighted sum of lot categories. S46
Q17	B	L1 × 1.00, L2 × 2.10, L3 × 2.90, L4 × 3.10, L5 × 3.90. S46
Q18	C	X = 6×1 + 2×2.1 + 1×2.9 + 1×3.1 = 16.2; LQI = 16.2 ÷ 10 = 1.62. S46
Q19	B	S45. S45
Q20	B	1 define criteria, 2 assign weights, 3 gather data, 4 evaluate, 5 calculate, 6 review, 7 communicate, 8 use for decisions. S39
Q21	D	Large quantities and amounts of money are involved. S33
Q22	A	Sharing POS data is a cure (S67). S61
Q23	A	Costs and lead time increase; availability and profitability decrease. S64
Q24	C	S64. S64
Q25	A	S66. S66
Q26	D	They give single-point control. S67
Q27	D	S68. S68
Q28	B	Also EDLP and tying promotions to sell-through. S69
Q29	A	S62. S62
Q30	A	SAP 22.5%, Oracle 16.1%, Salesforce 10.6%, Microsoft 6.4%, Amdocs 4.9%. S73
Q31	B	S72. S72
Q32	A	S87. S87
Q33	D	The fourth is learning and growth. S87
Q34	B	Stockout rate is its reverse. S84
Q35	B	End-to-end delay in a business process. S83
Q36	A	S80. S80
Q37	C	S88. S88
Q38	B	S95. S95
Q39	D	Also: no clear strategy, generic metrics, self-serving managers. S96
Q40	B	S89. S89
Q41	C	S109. S109
Q42	A	EV = PV × % complete. S111
Q43	A	EV = 1,00,000 × 0.6 = 60,000; CPI = 60,000 ÷ 50,000 = 1.2 (under budget). S111
Q44	C	CPI > 1 under budget; = 1 on budget. S111
Q45	B	Cost reduction uses the lowest price paid. S112
Q46	A	S112. S112
Q47	C	S120. S120
Q48	B	Asset efficiency = sales ÷ average asset value. S122
Q49	C	Gross Margin Return on Investment. S122

Q	ANS	WHY · SLIDE
Q50	C	Absolute deviation stops positive and negative errors cancelling. S116

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